

EXPERIMENT-9

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To design input forms that validate data such as email and phone number and display error messages using HTML, CSS, and JavaScript.

Form validation is used to ensure that the user inputs correct and meaningful data before submitting a form. It improves data accuracy and enhances user experience by providing immediate feedback. In this experiment, a form is created and validated using HTML, CSS, and JavaScript.

SETTING UP HTML FORM:

```
<!DOCTYPE html>
<html>
<head>
  <title>Form Validation</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

<div class="container">
  <h2>Registration Form</h2>

  <form id="myForm">

    <label>Email:</label>
```

```
<input type="text" id="email" placeholder="Enter email">
<small id="emailError"></small>

<label>Phone Number:</label>
<input type="text" id="phone" placeholder="Enter phone number">
<small id="phoneError"></small>

<button type="submit">Submit</button>

</form>
</div>

<script src="script.js"></script>

</body>
</html>
```

OUTPUT:

Registration Form

Email:	abc@gmail.com	Phone Number:	1234567890	<input type="submit" value="Submit"/>
--------	--	---------------	---	---------------------------------------

In this step, the basic structure of the form is created using HTML. Input fields are provided for email and phone number

STYLING USING CSS:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Form Validation</title>
```

```
<style>
```

```
  body {  
    font-family: Arial;  
    background-color: #f2f2f2;  
  }
```

```
  .container {  
    width: 300px;  
    margin: 100px auto;  
    padding: 20px;  
    background: white;  
    border-radius: 10px;  
  }
```

```
  h2 {  
    text-align: center;  
  }
```

```
  input {  
    width: 100%;  
    padding: 10px;  
    margin: 10px 0;  
  }
```

```
button {  
    width: 100%;  
    padding: 10px;  
    background-color: blue;  
    color: white;  
    border: none;  
    cursor: pointer;  
}
```

```
small {  
    color: red;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
  <h2>Registration Form</h2>
```

```
  <form id="myForm">
```

```
    <label>Email:</label>
```

```
    <input type="text" id="email" placeholder="Enter email">
```

```
    <small id="emailError"></small>
```

```
    <label>Phone Number:</label>
```

```
    <input type="text" id="phone" placeholder="Enter phone number">
```

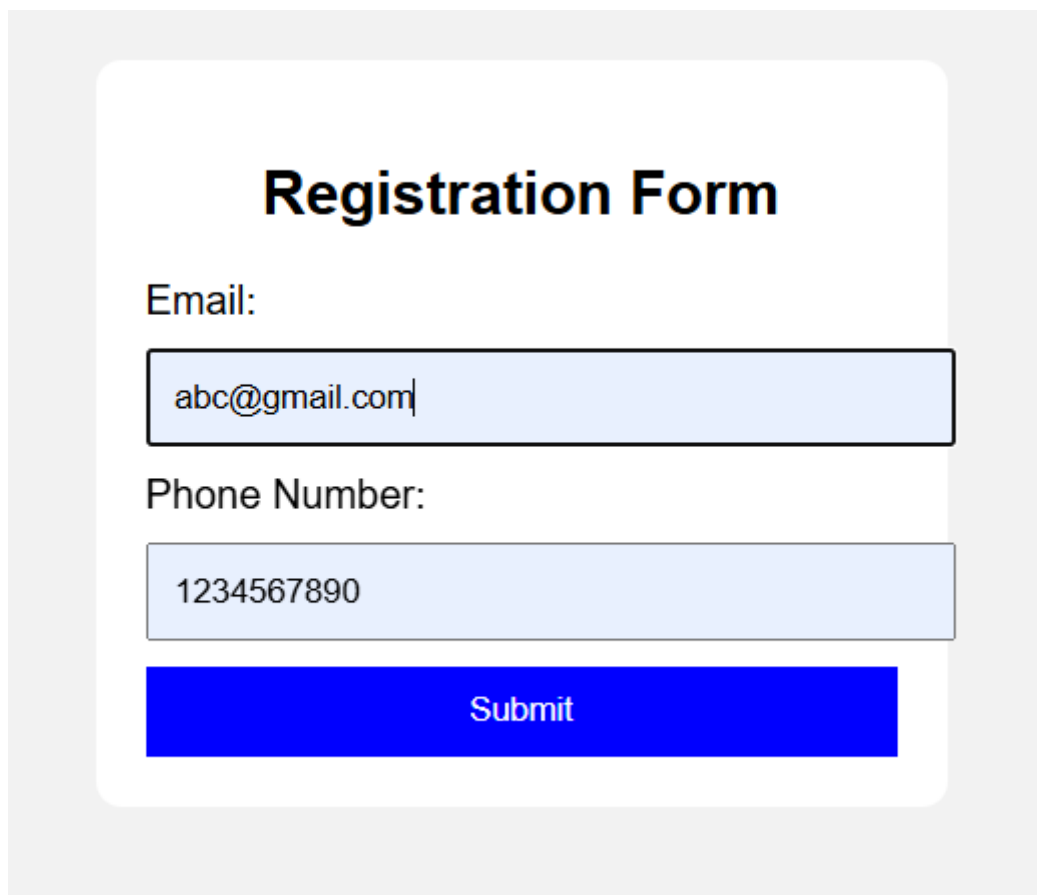
```
    <small id="phoneError"></small>
```

```
<button type="submit">Submit</button>

</form>
</div>

</body>
</html>
```

OUTPUT:

The image shows a registration form with a light gray background. The form itself is a white rounded rectangle. At the top, it has the title "Registration Form" in bold black text. Below the title, there are two input fields. The first is labeled "Email:" and contains the text "abc@gmail.com". The second is labeled "Phone Number:" and contains the text "1234567890". At the bottom of the form is a blue button with the text "Submit" in white.

CSS is used to enhance the visual appearance of the form by adding styles such as background color, spacing, alignment, and button design.

VALIDATION USING JAVASCRIPT:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Form Validation</title>
```

```
<style>
```

```
  body {  
    font-family: Arial;  
    background-color: #f2f2f2;  
  }
```

```
  .container {  
    width: 300px;  
    margin: 100px auto;  
    padding: 20px;  
    background: white;  
    border-radius: 10px;  
  }
```

```
  h2 {  
    text-align: center;  
  }
```

```
  input {  
    width: 100%;  
    padding: 10px;  
    margin: 10px 0;  
  }
```

```
  button {
```

```
width: 100%;  
padding: 10px;  
background-color: blue;  
color: white;  
border: none;  
cursor: pointer;  
}
```

```
small {  
  color: red;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
  <h2>Registration Form</h2>
```

```
  <form id="myForm">
```

```
    <label>Email:</label>
```

```
    <input type="text" id="email" placeholder="Enter email">
```

```
    <small id="emailError"></small>
```

```
    <label>Phone Number:</label>
```

```
    <input type="text" id="phone" placeholder="Enter phone number">
```

```
    <small id="phoneError"></small>
```

```
    <button type="submit">Submit</button>
```

```
</form>
```

</div>

<script>

```
document.getElementById("myForm").addEventListener("submit", function(e) {
```

```
    e.preventDefault();
```

```
    let email = document.getElementById("email").value;
```

```
    let phone = document.getElementById("phone").value;
```

```
    let emailError = document.getElementById("emailError");
```

```
    let phoneError = document.getElementById("phoneError");
```

```
    emailError.textContent = "";
```

```
    phoneError.textContent = "";
```

```
    let isValid = true;
```

```
    // Email validation
```

```
    if (!email.includes("@") || !email.includes(".")) {
```

```
        emailError.textContent = "Invalid email address";
```

```
        isValid = false;
```

```
    }
```

```
    // Phone validation
```

```
    if (phone.length !== 10 || isNaN(phone)) {
```

```
        phoneError.textContent = "Phone must be 10 digits";
```

```
        isValid = false;
```

```
    }
```

```
    if (isValid) {
```

```
        alert("Form submitted successfully!");
```

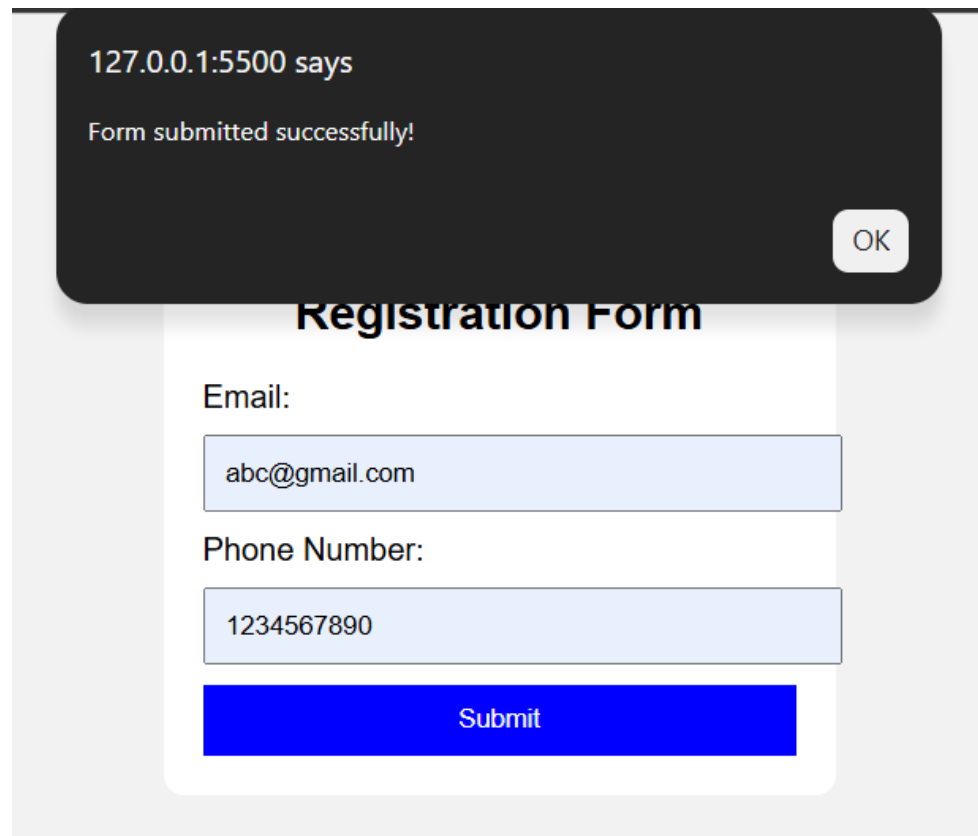
```
    }
```



```
});  
</script>
```

```
</body>  
</html>
```

OUTPUT:



The screenshot displays a web browser window. At the top, a dark grey notification box with rounded corners contains the text "127.0.0.1:5500 says" in white, followed by "Form submitted successfully!" in a smaller white font. An "OK" button is located in the bottom right corner of this box. Below the notification, the main content area has a light grey background. Centered in this area is a white box with rounded corners titled "Registration Form" in bold black text. Inside this box, there are two labels: "Email:" and "Phone Number:". Under "Email:", there is a light blue input field containing the text "abc@gmail.com". Under "Phone Number:", there is a light blue input field containing the text "1234567890". At the bottom of the white box is a solid blue button with the word "Submit" in white text.

Registration Form

Email:

Invalid email address Phone Number:

JavaScript is used to validate the input fields. It checks whether the email contains “@” and “.” and verifies that the phone number is exactly 10 digits. Error messages are displayed if validation fails.